

Artificial Intelligence Demo Document

Artificial Intelligence (AI) is the field of computer science that enables machines to perform tasks that normally require human intelligence.

Machine Learning (ML) is a subset of AI where systems learn patterns from data instead of being explicitly programmed.

Deep Learning is a subset of Machine Learning that uses neural networks with many layers.

Natural Language Processing (NLP) allows computers to understand and generate human language.

Computer Vision enables machines to analyze and understand images and videos.

Examples of AI applications include:

- Chatbots
- Recommendation systems
- Fraud detection
- Self-driving cars
- Medical diagnosis

Advantages:

- Automation
- Faster decision making
- Improved accuracy
- Better customer support

Challenges:

- Data privacy
- Bias in AI models
- High computational cost
- Ethical concerns